AI FOR DATA MANAGEMENT

PART 2: SYSTEM IMPLEMENTATION

TECHNISCHE UNIVERSITÄT DARMSTADT



PART 2: SYSTEM IMPLEMENTATION

You will implement a small **semantic data system on your own** and come up with your own **novel ideas to optimize its accuracy and execution costs**.

- We provide benchmark (data + queries), template code, and OpenAI access
- You will implement the system/operators to execute these (and other) queries
- You will optimize your system‘s accuracy and execution costs

You will present your system and ideas to us in a **15 minutes testat** (=oral) individually at the end of the semester (March 2027).

PART 2: SYSTEM IMPLEMENTATION

Part 2.1: Implement Query Execution (25% of grade)

- Execute previously-provided queries
- Execute held-out queries during testat

Part 2.2: Optimize Accuracy / Cost of your System (25% of grade)

- Integrate existing optimizations into your system (see paper presentations)
- Come up with novel optimizations and integrate them into your system
- Explain optimization during testat and show their effect

PART 2.1: IMPLEMENT QUERY EXECUTION

Goal is to implement a GENERAL semantic query processing engine

→ Not just for some specific queries

Supported Operators:

- Semantic Filter, Semantic Map, Semantic Join
- Relational Projection, Relational Filter, Relational Join, GROUP BY, ORDER BY, COUNT, LIMIT
- Must be composable into different query plans

Supported Modalities:

- Table, Text, Image
- All semantic operators must support all modalities, also cross-modality joins

Data: SemBench (Movie, Animals/Wildlife, MMQA, E-Commerce)

SEMBENCH

Benchmark for Semantic Query Processing Systems

Datasets + Queries + Evaluation Logic

Multiple Scenarios: Movie, Wildlife/Animals, E-Commerce, MMQA, Cars

Clone and execute repository to prepare the data

- Important parameter: `scaling_factor` (how many rows in main table, start with small values to manage cost)

- Website: https://sembench.github.io/SemBench/
- Paper: https://arxiv.org/abs/2511.01716
- Repo: https://github.com/SemBench/SemBench

QUERYING INTERFACE

You are free to design your own querying interface

Must be as convenient as possible to execute new queries

→ Little manual effort

Query Language (not recommended)

```
1  SELECT COUNT(*) AS positive_review_cnt
2  FROM Reviews R
3  WHERE R.id = 'taken_3'
4  AND AI.IF(R.reviewText, 'this review expresses a positive
       sentiment');
```

[Lao 2026]

Easier Option: Python API (highly recommended)

```
9   emails = pz.Dataset(source="enron-emails", schema=Email)
10  emails = emails.filter("The email is not quoting from a news
    ↪  article or an article ...")
11  emails = emails.filter("The email refers to a fraudulent scheme
    ↪  (i.e., 'Raptor', ...")
12
13  # user specified policy and plan execution
14  policy = pz.MinimizeCostAtFixedQuality(min_quality=0.8)
15  results = pz.Execute(emails, policy=policy)
```

[Liu 2025]

During testat, you will have 15 minutes to execute held-out queries with your system

Hint: Make your system integrate well with pandas or DuckDB for relational operators

LLM ACCESS

LLMs from OpenAI:

- Accessed through API proxy accessible from eduroam or VPN
- Monthly cost budget, contact us if used up
- You receive a secret user ID via Moodle which you must include in every request

Available Models:

- Text/images: `gpt-6-luna, gpt-6.1-terra, gpt-6.1-sol`
- Embeddings: `text-embedding-3-small, text-embedding-3-large`
- You can also use local models (e.g., SBERT for embeddings)

More details here: https://developers.openai.com/api/docs/models/compare

Hints:

- Start experimenting with small SemBench scaling factor to save costs
- See Example Python file in Moodle on how to use the API

PART 2.1 GRADING

1. Execute Previously-Provided Queries

- List of queries: see file on Moodle
- You must compute & show (plot/table) SemBench quality metrics (per query & per scenario)
- We will double-check actual query results during testat

2. Execute Held-Out Queries During Testat

- You will receive some held-out queries (in BigQuery syntax) for the four SemBench scenarios (Movie, Animals/Wildlife, MMQA, E-Commerce) that use operators from slide 4
- You will get some time to execute them with your system
- We will manually check query results during testat

PART 2.2: OPTIMIZE ACCURACY / COST

- Accuracy = SemBench quality metrics
- Cost = Cost for LLM & embedding requests

- Integrate **existing optimizations** into your system (see paper presentations)
- Come up with **novel optimizations** and integrate them into your system

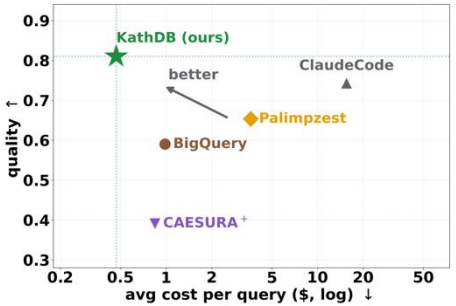
Potential Optimizations:

- Operator-level (do multiple operators)
- Plan-level

PART 2.2 GRADING

Your grade will depend on:

- How many optimizations you develop
- How novel they are
- How well they work



[Xiao 2026]

Also: How well can you explain them

Show cost-accuracy pareto plots for multiple queries

Important for Testat:

- Prepare some slides explaining your ideas (more details later)
- Be prepared to point at the relevant code

PROJECT SETUP / SUBMISSION

- You must submit your repository to Moodle as a ZIP archive before the testat
- You will set up your project in a clean (Docker) environment from your ZIP archive during the testat
- Do not include the SemBench data in the ZIP archive
  - We will provide the data (from Google Drive) during testat

SEMINAR ETHICS & RULES

Attendance (mandatory):

- Kickoff
- At least 4/5 student presentation sessions
- At least one office hour
- Slot of your final testat

If you cannot attend in person because of a semester abroad, please write us an email at the beginning of the semester.

Plagiarism:

- Implementation task is an individual contribution. Everyone must implement it independently.
- Do not share solutions with other students
- We check all code submitted for grading with plagiarism detection software

AI Assistance:

- Allowed, for example for coding
- You will be graded based on your understanding of the topic
- We will check this in the testat

Violating these rules counts as academic misconduct and will be subject to disciplinary action (2 x misconduct → you might fail your degree program)

SEMINAR TIMELINE

Schedule

Tue, 13.10.2026	09:50-11:30	S306/146	Kickoff
Tue, 20.10.2026	09:50-11:30	S306/146	Lecture: LLMs & Databases
Tue, 27.10.2026	09:50-11:30	S306/146	Lecture: Paper Reading & Presentation Tips
Tue, 03.11.2026	09:50-11:30	S306/146	Lecture: Semantic Data Systems & Instructions for Part 2
Tue, 10.11.2026	09:50-11:30	S306/146	Student Presentations: Semantic Query Processing Engines I
Tue, 17.11.2026	09:50-11:30	S306/146	Student Presentations: Semantic Query Processing Engines II
Tue, 24.11.2026	09:50-11:30	***S103/121***	Student Presentations: Plan-level Optimizations
Tue, 01.12.2026	09:50-11:30	S306/146	Student Presentations: Operator-level Optimizations
Tue, 08.12.2026	09:50-11:30	S306/146	Student Presentations: Benchmarks and Agent-first Systems & Details on Testats
Tue, until end of semester	09:50-11:30	S202/A213	Office Hours
10.-12.03.2027	09:00-14:00	S202/A213	Final Testats (Details will follow)

LECTURE NEXT WEEK

Topic: Student Presentation Session

Tuesday, 10.11.2026, 09:50-11:30, Room S3 06|146

QUESTIONS

